

A SAMPLE MATHEMATICAL NOTE

YOUR NAME

ABSTRACT. A small example of shared macros, mathematical notation, and theorem environments. Edit this file for the content and macros.tex for notation.

1. NOTATION

We write $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ for the usual number systems. Let $\mathcal{H}$ be a real inner product space, with norm $\|u\| = \sqrt{\langle u, u \rangle}$. For example, a set and an automatically sized expression are

$$B := \{x \in \mathbb{R} \mid |x| < 1\}, \quad \left\| \frac{u+v}{2} \right\| \leq \frac{\|u\| + \|v\|}{2}.$$

For a random variable X , expectation, probability, and variance are written $\mathbb{E}X$, $\mathbb{P}(X > 0)$, and $\text{Var}(X)$.

Son: Adjust the notation to match your paper.

2. A THEOREM AND A PROOF

Theorem 2.1 (Cauchy–Schwarz inequality). *For all $u, v \in \mathcal{H}$,*

$$(2.1) \quad |\langle u, v \rangle| \leq \|u\| \|v\|.$$

Proof. If $v = 0$, the claim is immediate. Otherwise, set $t = \langle u, v \rangle / \|v\|^2$. Expanding a square gives

$$0 \leq \|u - tv\|^2 = \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2}.$$

Rearranging proves (2.1). $\square$

Remark 2.2. Theorem 2.1 illustrates labels and cross references. Lemmas, propositions, corollaries, definitions, and examples are also available and share the theorem counter within each section.

3. CALCULUS

The shared macros also cover derivatives and vector calculus:

$$\frac{d}{dt} \int_{\mathbb{R}} f(t, x) \, dx = \int_{\mathbb{R}} \frac{\partial f}{\partial t}(t, x) \, dx, \quad \Delta u = \nabla \cdot (\nabla u).$$

The displayed identity assumes enough regularity to differentiate under the integral. Higher derivatives use `\pdv[2]{u}{x}`, giving $\frac{\partial^2 u}{\partial x^2}$.

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